Team Falcons at ImageEval 2026 Shared Tasks: Structure-Constrained QLoRA Fine-Tuning and Dual-Model Consensus Verification

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Abstract

We present the Team Falcons system for the ImageEval 2026 Task 1b English track. The task provides culturally grounded images from the OASIS/EverydayMMQA resource family and three English statements per image, exactly one of which is grounded in the image (Abdaljalil et al., 2026; Alam et al., 2025; Mousi et al., 2026b). Rather than treating the statements as independent binary decisions, we exploit this structural constraint and formulate each item as image-conditioned three-way classification. In Stage 1, we fine-tune Qwen3-VL-8B (Bai et al., 2025) with 4-bit QLoRA (Dettmers et al., 2023; Hu et al., 2022) and estimate uncertainty from the first-token probabilities assigned to A, B, and C. In Stage 2, we send only examples whose top-two probability margin is below 0.90 to Gemini 3.6 Flash and GPT-5.4-mini. Their answer replaces the Stage 1 prediction only when both verifiers return the same valid option. On the official test set, Stage 1 obtains a Contrastive Instability (CI) of 0.0360 and Combined Accuracy of 0.9640. Consensus verification improves these scores to 0.0290 and 0.9710, respectively, placing our submission first on the Task 1b English leaderboard. After test labels were released, we found that the verifiers agreed on 91 of 138 routed items and changed 17 predictions: 12 corrections and 5 harmful revisions. The gain over Stage 1 is not statistically significant under a two-sided exact McNemar test ($p = 0.1435$). The system illustrates selective combination of a local open-weight VLM with proprietary verifiers, without establishing an optimal routing policy.

1 Introduction

Vision-language models (VLMs) have advanced rapidly on visual question answering, yet their outputs may still mention objects, attributes, or relations that are unsupported by the image (Liu et al., 2024; Li et al., 2023). Such errors are particularly subtle in culturally grounded settings: a statement may be plausible under general or regional knowledge while contradicting the visual evidence. M2CQA shows that a model can identify a true statement and still accept a paired counterfactual, with higher counterfactual hallucination rates in Arabic and its dialects (Mousi et al., 2026b).

ImageEval 2026 evaluates culturally grounded multimodal capabilities for Arabic contexts (Abdaljalil et al., 2026). We participated only in Task 1b English, where each item contains an image and three statements, exactly one of which is true. The official ranking metric is Contrastive Instability (CI; lower is better) (Mousi et al., 2026a). If a system always outputs one `true` and two `false` labels, the task reduces to predicting the unique true statement. This formulation prevents logically inconsistent outputs and aligns the training objective with item-level correctness.

Our contributions are threefold. First, we formulate Task 1b as constraint-preserving A/B/C classification and fine-tune an open-weight VLM accordingly. Second, we route low-margin items to two separately queried multimodal verifiers and require strict consensus before changing the local model's prediction. Third, we use official results and post-hoc analyses with the released test labels to quantify each stage and characterize the remaining errors. The final system achieves a CI of 0.0290 and ranks first in Task 1b English.

2 Task and Related Work

2.1 Task and Metrics

The data belongs to the OASIS/EverydayMMQA resource family, which covers everyday cultural settings across 18 Arab countries and provides multimodal content in English and Arabic (Alam et al., 2025). Task 1b English contains 3,000 training items, 500 development items, and 1,000

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blind test items. Each item has a label vector $\mathbf{y} \in \{0,1\}^3$ with $\sum_i y_i = 1$, although submissions use a separate True/False prediction for each statement.

CI is the fraction of incompletely correct items among items with at least one correct statement label. Combined Accuracy requires all three labels to be correct. Q+ is accuracy on the unique gold-true statement, and Q- is accuracy over the two gold-false statements per item. Our output is always one-hot: choosing the wrong option still correctly rejects one false statement. Thus, writing item-level accuracy as a, our system satisfies

$$\begin{aligned} \text{Combined} = Q^+ &= a, \\ Q^- &= \frac{1+a}{2}, \quad \text{CI} = 1-a. \end{aligned} \tag{1}$$

These identities depend on both gold and predicted labels being one-hot. Counterfactual Hallucination Rate (CFHR) measures incorrect acceptance of a false statement conditional on correctly accepting the true statement; it is necessarily zero for our output format, rather than independent evidence of hallucination-free reasoning.

2.2 Visual Hallucination and Efficient Adaptation

POPE casts a VLM's tendency to affirm nonexistent objects as a discriminative evaluation task (Li et al., 2023). Subsequent surveys attribute hallucinations to the training data, visual encoder, cross-modal alignment, and decoding process (Liu et al., 2024). Unlike generic object-presence benchmarks, ImageEval counterfactuals may differ only in material, activity, national association, or traditional use. Resolving them requires both fine-grained visual evidence and cultural knowledge.

We select Qwen3-VL for its dynamic-resolution visual processing and image-understanding capabilities (Bai et al., 2025). The open-weight 8B checkpoint supports local adaptation and inference within our single-GPU setup. This is a practical compatibility and resource rationale, not evidence that this backbone outperforms alternatives. LoRA reduces the number of trainable parameters through low-rank weight updates (Hu et al., 2022), while QLoRA trains adapters through a frozen 4-bit backbone to reduce memory use (Dettmers et al., 2023). We reserve proprietary models for selective verification rather than invoking them on every item.

3 System Description

3.1 Joint Three-Way Fine-Tuning

Given an image I and three statements s_A, s_B, and s_C, the prompt states that exactly one statement is grounded in the image and requires the model to return only A, B, or C. The target is the letter corresponding to the supported statement. We fine-tune Qwen3-VL-8B-Instruct with 4-bit QLoRA using Unsloth. We freeze the vision layers and inject LoRA adapters into the attention and MLP modules of the language model, with rank $r = 16$, scaling factor $\alpha = 32$, and dropout 0.

We train for one epoch on the official English training set with a learning rate of 2×10^{-4}, a per-device batch size of 2, and 8 gradient-accumulation steps, yielding an effective batch size of 16. We use 8-bit AdamW, cosine learning-rate scheduling, a warmup ratio of 0.03, and weight decay of 0.01. The maximum sequence length is 2,048, and the maximum image-pixel budget is $1024 \times 28 \times 28$. The submitted configuration uses neither option-order augmentation nor parallel Modern Standard Arabic (MSA) data.

At inference time, we use greedy decoding with vLLM (Kwon et al., 2023). We collect the first-token log probabilities for the candidate letters and normalize them over A/B/C to obtain p_A, p_B, and p_C. The predicted option is $\hat{i} = \arg\max_i p_i$. We convert it to the official format by assigning `true` to $\hat{i}$ and `false` to the other two options.

3.2 Low-Margin Dual-Model Verification

We define the Stage 1 probability margin as the difference between the largest and second-largest candidate probabilities:

$$m = p_{(1)} - p_{(2)}. \tag{2}$$

Stage 2 is triggered only when $m < \tau$; the submitted configuration uses $\tau = 0.90$. This is a heuristic routing cutoff, not a calibrated probability-of-correctness threshold or an established optimum. A higher cutoff admits more items for verification, while the separate consensus gate restricts overrides. We do not have a documented controlled development-set threshold sweep establishing the superiority of 0.90. The normalized first-token scores are used only as a routing signal; no temperature scaling or calibration assessment was performed.

Each verifier receives the original image and all three statements and returns one option with a brief

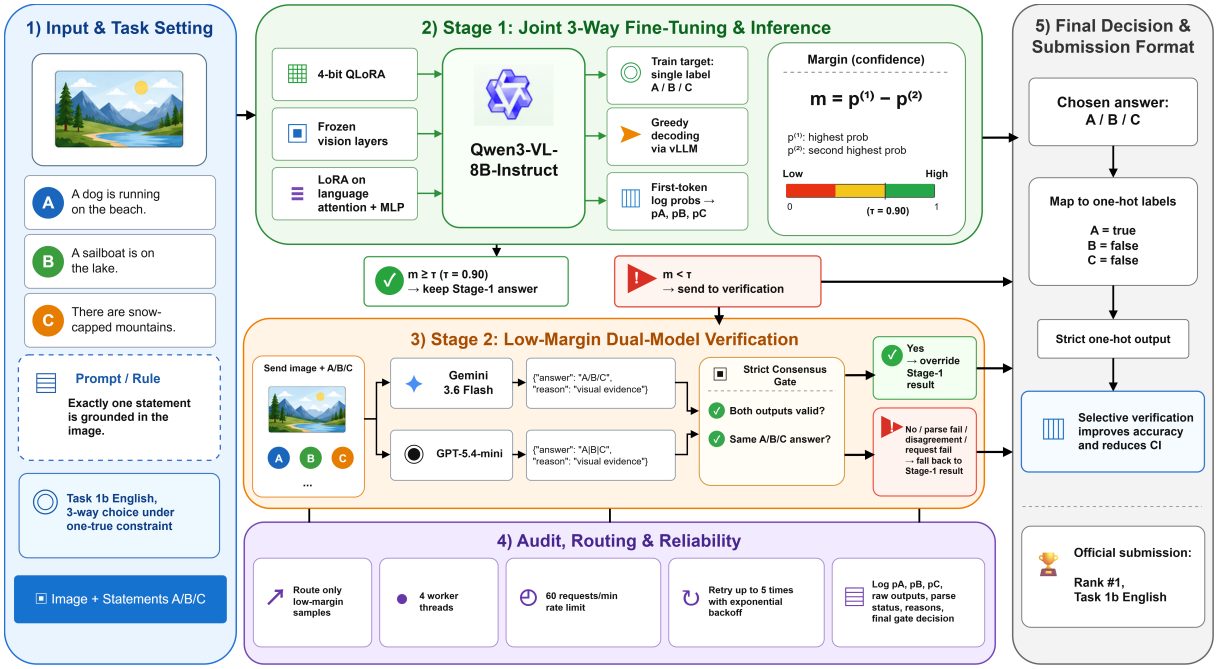


Figure 1: Overview of the submitted system. Stage 2 processes only low-margin items and changes the Stage 1 prediction only under strict agreement between the two verifiers.

visual justification in JSON. We use relay endpoints with the request identifiers `gemini-3.6-flash` and `gpt-5.4-mini`; these identify the requested services, not independently verified backend checkpoint versions. The client explicitly sends `temperature=0` to Gemini but omits the temperature field for GPT. It also leaves `top_p`, seed, and output-token limits unspecified for both verifiers. The retained code establishes this asymmetry but not its original rationale or the relay's effective GPT decoding settings. Therefore, matched deterministic decoding and repeatable consensus decisions cannot be assumed. Appendix A gives the complete application-level prompts.

The gate is deliberately conservative. A verifier answer overrides Stage 1 only when both models return valid and identical A/B/C predictions. If they disagree, fail to parse, or fail to respond, the system retains the fine-tuned model's prediction. This rule limits the effect of errors from either verifier and preserves a strict one-hot output for every item.

Figure 1 summarizes the complete data flow. Margin-based routing sends 138 of the 1,000 test items to Stage 2, for 276 verifier requests in total. The implementation uses four worker threads and a global rate limit of 60 requests per minute. Failed requests are retried up to five times with exponential backoff. For auditing, we store the Stage 1 probabilities, both raw verifier outputs, parsing status, justifications, the gating decision, and whether the final prediction differs from Stage 1.

System	CI↓	Comb.↑	Q+↑	Q-↑
Fine-tuned model	.0360	.9640	.9640	.9820
+ dual-model verification	**.0290**	**.9710**	**.9710**	**.9855**

Table 1: Official Task 1b English test results. The final system achieves CFHR = 0 and ranks first on the leaderboard.

4 Experiments and Results

4.1 Experimental Setup

We use the development set to select the training configuration and inspect errors. The official test set was unlabeled during the competition. After the organizers released the test labels, we evaluated both stages with the official scorer; the final scores exactly match the public leaderboard. Training and Stage 1 inference use the local Qwen3-VL-8B model, whereas Stage 2 uses external endpoints. Before submission, we validate the output schema, complete ID coverage, absence of duplicate rows, and exactly one `true` label per item.

4.2 Main Results and Ablations

Table 1 shows that dual-model verification reduces CI by 0.007 and increases Combined Accuracy from 0.9640 to 0.9710. Because our system always pro-

Override policy	Changes	CI↓	Comb.↑
No verifier	0	.036	.964
Gemini only	34	**.028**	**.972**
GPT only	50	.056	.944
Dual-model consensus (official)	17	.029	.971

Table 2: Post-hoc ablation using released test labels. Single-verifier rows were not official submissions.

duces strict one-hot outputs, Q+ equals Combined Accuracy and CFHR is 0. The Stage 1 row is a fine-tuned model, not an untuned-backbone baseline. To measure the contribution of adaptation, we subsequently ran a controlled 500-item development-set comparison using the same Qwen3-VL-8B checkpoint, prompt, image preprocessing, vLLM decoding parameters, and hardware, with the LoRA adapter as the only changed component. Appendix Table 6 reports that the untuned model obtains CI 0.046 and Combined Accuracy 0.954, while the adapter obtains CI 0.026 and Combined Accuracy 0.974. The paired comparison has 14 wrong-to-correct and 4 correct-to-wrong changes (two-sided exact McNemar $p = 0.0309$). This is one post-hoc development run, not a multi-seed estimate or a comparison to other backbones.

Of the 1,000 test items, 138 fall below the margin threshold. The two verifiers agree on 91 items and disagree on 47; every response is parsed successfully. Consensus matches Stage 1 on 74 items and changes 17. Evaluation against the subsequently released labels shows 12 wrong-to-correct and 5 correct-to-wrong changes, for a net gain of seven items. This decomposition matches the 0.7 percentage-point increase in Combined Accuracy. Thus, although the high threshold routes a relatively broad subset, consensus limits actual changes to 1.7% of the test set.

Table 2 isolates the effect of the consensus gate. Allowing either verifier to override Stage 1 independently yields sharply different outcomes: Gemini alone obtains CI 0.028, whereas GPT alone degrades CI to 0.056. The official submission requires dual-model consensus and therefore does not depend on post-hoc selection of the better verifier, while retaining most of the net gain. The single-verifier scores were computed only after test-label release and provide no evidence for model selection during the competition.

For the submitted system versus Stage 1, a two-sided exact McNemar test on paired item correctness uses 12 corrections and 5 harmful changes, yielding $p = 0.1435$. The observed gain is therefore not statistically significant at the 0.05 level. This post-hoc test treats items as independent units and does not measure training-seed or API-response variability; the leaderboard rank is descriptive, not evidence of a statistically reliable advantage.

Appendix Table 4 reports threshold sensitivity and request counts. At $\tau = 0.50$, 52 items require 104 verifier requests, compared with 138 items and 276 requests at 0.90. Both yield 0.971 accuracy: the lower threshold saves 172 requests (62.3%) in this replay. Thus, these data do not justify preferring 0.90 on an accuracy–request-count trade-off. The analysis reuses stored responses only for items queried during the competition and cannot evaluate thresholds above 0.90 or fresh API-response variability. Test labels were not used to choose the submitted threshold; a new deployment should select its cutoff on development data under an explicit request budget.

We additionally conducted a post-hoc development-set threshold analysis using a retained 500-item Stage 1 output (Combined Accuracy 0.970) and 41 Gemini/GPT response pairs collected for items with $m < 0.90$. Some Gemini responses contained explanatory prose followed by a valid JSON object; we replayed the saved responses after extracting the final valid JSON object, without further API calls. For each $\tau \leq 0.90$, items with $m \geq \tau$ retain the Stage 1 prediction. Appendix Table 5 reports every threshold from 0.10 to 0.90. Thresholds from 0.20 to 0.50 match the $\tau = 0.90$ Combined Accuracy of 0.980 using 20–30 rather than 82 requests. In this post-hoc replay, $\tau = 0.60$ obtains the highest Combined Accuracy, 0.982, with 44 requests. Therefore, neither this development analysis nor the test replay supports claiming that the submitted $\tau = 0.90$ is optimal. These thresholds were not available when the official system was submitted, and the results do not estimate variation from new verifier calls.

4.3 Error Analysis

Dual-model consensus nevertheless produces five harmful revisions on the official test set, suggesting that the proprietary models may share cultural priors or visual blind spots. Agreement does not imply correctness. Future systems should prospectively choose a cutoff on a held-out development set under an explicit request budget and use targeted crops or higher-resolution re-examination for material, texture, and OCR-related items. More-

Final test error category	Errors	Size	Error (%)
Food & Cooking	6	93	6.45
Culture, Arts & Entertainment	5	175	2.86
Vehicles & Transportation	4	148	2.70
Sports & Recreation	4	101	3.96
People, Society & Education	3	185	1.62
History/Geography/Identity	3	64	4.69
Religion & Spirituality	2	87	2.30
Other	2	147	1.36
Total	29	1,000	2.90

Table 3: Final-system errors using released test labels. Error rate is errors divided by category size; these are descriptive comparisons.

over, our probabilities come from one generated token without temperature scaling and should not be interpreted as calibrated posterior probabilities.

Table 3 distributes the final 29 errors across coarse categories. Food & Cooking has the highest error rate (6/93); the Beverages subcategory alone accounts for 5/34 errors (14.7%). Typical confusions include lemon slices versus cinnamon sticks, tacos versus pizza, and bread versus cured meat. Transportation errors include train station versus airport terminal and freight versus tourism uses. Culture and arts errors cover instruments, craft materials, and film or television scenes. In a complementary manual taxonomy, fine-grained food or object distinctions and color or material distinctions account for five errors each, while activities and venues account for three each.

Appendix Table 7 illustrates both outcomes of consensus verification. The statements are reproduced verbatim from the dataset, without translation or paraphrase. Successful revisions rely on specific visual evidence, such as Roman architectural features or the grooved structure of a honey dipper; harmful revisions instead follow general priors about embroidery materials or container uses. Training uses a single RTX 4090 with Unsloth, and Stage 1 inference uses vLLM. Appendix Table 8 summarizes the available resource information.

Category differences are exploratory, not demonstrated population-level effects. For example, the 95% Wilson interval for the Food & Cooking error rate is 2.99–13.37%, and that for History/Geography/Identity is 1.61–12.90%. Small error counts give substantial uncertainty; no category-wise significance claim is made.

5 Conclusion

We present Team Falcons' first-ranked system for ImageEval 2026 Task 1b English. The system reformulates independent statement judgments as constraint-preserving three-way classification, routes low-margin items to two proprietary multimodal verifiers, and requires strict consensus before overriding the local model. Verification reduces Contrastive Instability from 0.0360 to 0.0290 and raises Combined Accuracy from 0.9640 to 0.9710. The remaining errors show that culturally grounded hallucination detection requires both fine-grained visual discrimination and cultural knowledge. Future work should investigate probability calibration, targeted visual re-examination, and error-aware routing.

Limitations

We participated only in the English track, so our findings may not generalize to MSA or Arabic dialects. Stage 2 relies on proprietary endpoints whose weights, service versions, and default decoding settings are outside our control, and it incurs additional cost relative to a fully local system. Fine-grained verifier analyses use labels released after the competition; these labels did not influence the official submission. The reported model identifiers and parameters reflect the runtime configuration used in August 2026.

The method assumes exactly one grounded statement. If none or multiple statements are grounded, the forced A/B/C output cannot represent the correct label set, and the metric identities in Section 2.1 no longer apply. Extending the system would require independent statement labels or a structured multi-label output, including the empty set, together with revised training targets, routing scores, and a label-set consensus rule. We have not evaluated this extension. We have only one controlled development-set comparison of the untuned checkpoint and its adapter, no comparison to alternative backbones, no calibrated routing evaluation, and no repeated-run estimates of uncertainty. Consequently, we do not generalize the observed adapter gain beyond this setting or claim that the routing policy is optimal.

Ethics Statement

The system uses research data provided by the shared task and collects no additional personal data. Its cultural judgments may inherit regional biases

from the training data and proprietary models. It should therefore not be used for high-stakes decisions about individuals, social groups, or religious and cultural practices. Data and model use must comply with the applicable licenses and terms of service.

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A Prompts and Reproducibility Details

The following is the complete Stage 1 text template used for training and inference. Braced placeholders are replaced by the original statements in dataset order, not omitted instructions.

```
Exactly one statement is grounded in the image.

A. {a}
B. {b}
C. {c}

Return only A, B, or C.
```

The user message contains the image followed by this text. Training adds an assistant message containing the target letter. The model processor applies its chat template; the application supplies no explicit system message.

Both Stage 2 verifiers receive the following text in a single user message, followed by the original image as an image-URL content part. No application-level system message or Stage 1 answer is sent.

```
You are verifying a three-choice visual question.

Exactly one of statements A, B, and C is true.
    Inspect the image carefully and
select the true statement. Do not rely on the
    wording alone.

{metadata}Statements:
A. {statement_a}
B. {statement_b}
C. {statement_c}

Return JSON only, with no Markdown fences:
{"answer":"A|B|C","reason":"brief visual
    evidence"}
```

When metadata inclusion is enabled, nonempty Country, Category, and Subcategory fields are serialized in that order as `Field: value` lines followed by a blank line. For the retained blind-test inputs these fields are absent, so `{metadata}` expands to the empty string; metadata added with released gold labels is not retroactively included in verifier requests. The displayed JSON braces are literal after Python template formatting. Service-side system prompts are not available to the authors. Our training seed is 2026; this does not imply a fixed API sampling seed.

B Supplementary Experiments

τ	Routed	Requests	Changes	CI↓	Comb.↑
.10	6	12	0	.036	.964
.20	21	42	3	.033	.967
.30	29	58	5	.033	.967
.50	52	104	9	.029	.971
.70	80	160	12	.030	.970
.90	138	276	17	.029	.971

Table 4: Post-hoc sensitivity of dual-model consensus to the margin threshold. Test labels were not used to tune the submitted threshold.

τ	Routed	Requests	Changes	CI↓	Comb.↑
.10	1	2	1	.028	.972
.20	10	20	5	.020	.980
.30	11	22	5	.020	.980
.40	12	24	5	.020	.980
.50	15	30	5	.020	.980
.60	22	44	6	**.018**	**.982**
.70	28	56	7	.020	.980
.80	32	64	7	.020	.980
.90	41	82	7	.020	.980

Table 5: Post-hoc development-set threshold analysis. All rows replay the 41 saved response pairs collected at $m < .90$; lower thresholds retain the Stage 1 prediction for remaining items. The results were unavailable for selecting the official submission.

C Resource Use

Supervised fine-tuning uses only the official English training set. The retained API audit covers 138 items and 276 requests but does not include server-side token usage. For token-based proxy pricing, total cost would be $\sum_k (n_k^{\text{in}} c_k^{\text{in}} + n_k^{\text{out}} c_k^{\text{out}})/10^6$. Image-token counts require proxy billing records and cannot be reconstructed reliably from request counts alone.

System	CI↓	Comb.↑	Q+↑	Q-↑
Qwen3-VL-8B, no adapter	.046	.954	.954	.977
Qwen3-VL-8B + QLoRA adapter	**.026**	**.974**	**.974**	**.987**

Table 6: Controlled 500-item English development comparison. Both rows use the same local Qwen3-VL-8B checkpoint, prompt, image preprocessing, vLLM decoding parameters, and RTX 4080 SUPER; the adapter is the only changed component. The paired comparison has 14 wrong-to-correct and 4 correct-to-wrong changes (two-sided exact McNemar $p = .0309$). This post-hoc single-seed result is not a test-set result or a multi-seed estimate.

Outcome	Scenario	Original English statements (verbatim)	Decisions
Corrected	Tunisia / temple	**A.** The temple was most likely designed by the Ancient Greek civilization. True or False? **B.** The temple was most likely designed by the Ancient Egyptian civilization. True or False? **C.** The temple was most likely designed by the Ancient Roman civilization. True or False?	Stage 1: A Consensus: C Gold: C
Corrected	Kuwait / honey	**A.** The primary use of the wooden tool resting on the jar is to extract honey from the jar. True or False? **B.** The primary use of the wooden tool resting on the jar is to stir honey into tea or dishes. True or False? **C.** The primary use of the wooden tool resting on the jar is to spread honey on bread. True or False?	Stage 1: B Consensus: A Gold: A
Harmful	Syria / embroidery	**A.** Silk thread is likely used to create the embroidery on this traditional dress. True or false? **B.** Cotton thread is likely used to create the embroidery on this traditional dress. True or false? **C.** Metal wire is likely used to create the embroidery on this traditional dress. True or false?	Stage 1: B Consensus: A Gold: B
Harmful	Yemen / drink	**A.** The primary purpose of the white ceramic bowl in the image is to serve as a decorative base for the glass cup. True or False? **B.** The primary purpose of the white ceramic bowl in the image is to hold liquid. True or False? **C.** The primary purpose of the white ceramic bowl in the image is to cook food. True or False?	Stage 1: A Consensus: B Gold: A

Table 7: Representative prediction changes. Statements are reproduced verbatim from the test set and inference audit files. The first two are successful corrections; the latter two are harmful revisions.

Resource	Configuration
Training GPU	NVIDIA GeForce RTX 4090
Peak GPU memory	8 GB
Training time	30 mins
Stage 1 inference time	4 mins
API requests	276
API cost	USD 3.50
PyTorch/Transformers	2.10 / 5.5.0
Training/inference frameworks	Unsloth / vLLM
Random seed	2026

Table 8: Compute and software resources.